Sahaj Bhadja

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Summary

Associate Software Engineer with 2+ years of experience building **TypeScript/React experiences**, **low-latency transactional services**, and user-centered AI workflows across cloud-based applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ consistency across 10K+ submissions through **structured model evaluation**, persistent results, fallback paths, and reviewer controls supporting reproducible testing and responsive user-facing analytical workflows reliably at scale.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Increased lead conversions by 15% through **customer-centered workflow enhancements**, implementing role-based access, real-time updates, notifications, automated tests, monitoring, and reliable Docker/Jenkins deployments across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Reduced manual refund handling by approximately 70% through **payment workflow automation**, supporting refunds, disputes, and consistent status across React Native, TypeScript, Android, iOS, and web experiences reliably.

Skills

Languages: TypeScript, JavaScript, Java, Python, SQL, HTML, CSS
Frontend & UX: React, React Native, Responsive Design, Reusable Components, User-Centered Design, Material UI, Tailwind CSS
Backend & Transactions: Node.js, FastAPI, Flask, REST APIs, Microservices, WebSockets, Payment Webhooks, Transactions, Caching
AI & Data: LLMs, Transformers, PyTorch, RAG, FAISS, Pinecone, Model Evaluation, PostgreSQL, MySQL, MongoDB
Cloud & Quality: AWS, GCP, Docker, Kubernetes, Git, GitHub, Jenkins, CI/CD, Unit Testing, Integration Testing, Automated Testing, PyTest, Jest, Debugging, Observability

Projects

Conversational Travel Planning Experience - Design Concept
- Designed a **conversational travel-planning experience** concept combining React interfaces, TypeScript services, model-provider APIs, preference-aware recommendations, streamed responses, confirmation checkpoints, evaluation datasets, fallback handling, and booking-context preservation reliably.

Low-Latency Booking Orchestrator - Design Concept
- Designed a **low-latency booking orchestrator** concept coordinating inventory, pricing, reservations, payments, cancellations, and refunds through idempotent APIs, cached queries, transactional state transitions, performance tests, and observable recovery workflows.

Big-Brain | Live
- Built a **retrieval-augmented generation platform** using LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, and REST APIs, sustaining 95% query accuracy while enabling real-time document analysis reliably for 200+ users.